Towards a Cartography of Datacentering

A Multimodal Research Agenda for Innovation Studies

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Today

1. **The empirical moment** — why data centers land in Research Policy right now
2. **The gap** — what innovation studies is not yet asking
3. **An arena of transformation** — theoretical + methodological framework
4. **First results** — 40,824 articles: register shift, topic clusters, actor network
5. **Three contributions** — Cartalog of Actors, Cartalog of Events, open toolkit

The Empirical Moment

Metric	Value
Hyperscaler capex 2026	**$750B**
DC electricity consumption 2025	485 TWh (+17% YoY)
Data center demand growth 2025–26	+50% YoY (Goldman Sachs 2026)
Hyperscaler share of DC capacity 2031	67% of global total

"The largest concentrated infrastructure build of our lifetimes — happening now."

Not a future scenario. Not a policy proposal.

An arena in formation.

Datacentering: the contested, multi-scalar process by which data center infrastructure becomes embedded in energy systems, planning regimes, financial markets, and built environments — reshaping each in turn.

CreditSights (Feb 2026);
IEA Energy & AI (Apr 2026);
Synergy Research Group (2025)

Why Now? The Contested Arena

- **Stargate:** $500B / ~7 GW committed (OpenAI, Oracle, SoftBank)
- **PJM electricity:** capacity prices **+833%** (2024/25→2025/26); DC = 40% of auction cost [PJM Dec 2025]
- **Maine:** legislature passes statewide moratorium bill — governor vetoes, Apr 2026 [DCD corpus; Mills veto 24 Apr 2026]
- **Loudoun Co.:** eliminates by-right DC zoning, March 2025 [DCD corpus]
- **DeepSeek:** re-opens Jevons paradox debate on AI compute demand[a]

37,552 DCD articles

2008–2026, headline + intro

1,073 contestation events

Peak contestation: 2025

Trade press = primary event record before academic literature

[a] Jevons paradox: efficiency gains increase, not decrease, total consumption by stimulating demand.

The Gap in Innovation Studies

RP canon covers:

Platforms (Gawer 2014)
GPTs (Brynjolfsson)
Sectoral systems (Malerba)
Transitions (Geels)

Critical DC studies exists:

Big Data & Society, *Convergence*
Edwards et al. 2025; Monstadt 2025
— but outside RP's canon

Missing:

Integrative framework for the *infrastructural + methodological* turn in AI-era innovation studies

No unified approach to:
- mapping contested arenas
- surfacing implicated actors
- operationalizing multimodal evidence

Theory: Arenas of Transformation

After Jørgensen's *arenas of development* (2012, *Res. Policy*)

Flat ontology. Start from *actor constellations* and their collective engagement — not structural tiers.

Negotiation is the fundamental process.

Goal: **characterize what is *emerging*** before the arena stabilises.

+ Clarke (2005): situational maps

Implicated actors: silenced (communities, visa workers) + constructed but absent ("the consumer", "future generations")

The datacentering arena (in formation):

Co-constitutive.
Not hierarchically ordered.

Methodological Framework: Issue Mapping + CITC

Marres (2015) — *ST&HV*

Issue mapping via digital traces.
Trade press, hyperlinks, hashtags as empirical window into issue formation.

Berente et al. (2019); Miranda et al. (2016)

Computationally Intensive Theory Construction

Sampling → synchronic → diachronic → **lexical framing**

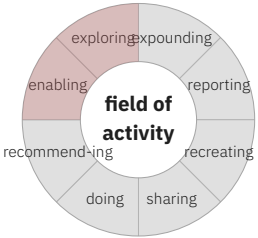
Patterns become contributions *only* once situated in scholarly discourse.

Marres tells us
what to trace

CITC tells us
how to trace it
rigorously enough for RP

Prevents atheoretical pattern-mining.

Multimodal Registerial Cartography



contestation register

Matthiessen (2015): registerial cartography

We extend to three registers:

Verbal	Hallidayan process types — verb inventory of headlines + intros
Visual	CLIP embeddings + Jancsary ideational / interpersonal coding of 34,869 images
Spatio-relational	city2graph urban embeddedness — DC node in power, residential, commercial graph

Verbal register — DCD corpus:

Contestation: *enabling* + *exploring* ↑

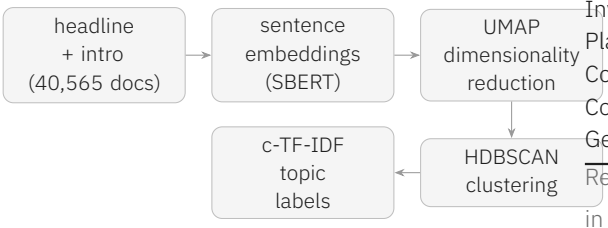
Non-contestation: *reporting* + *doing*

BERTopic: What We Run It On

Input: 40,565 headlines + intros

From `datacentering_crawl_dcd.tsv`:

"Microsoft expands plans for La Porte, Indiana, data center campus. Company seeks to rezone..."



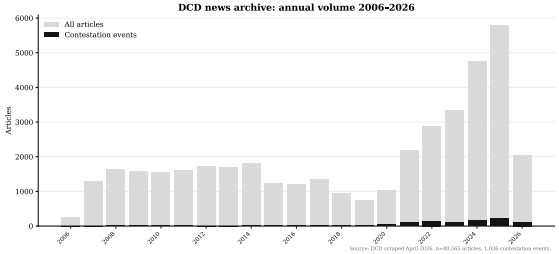
Expected topic clusters

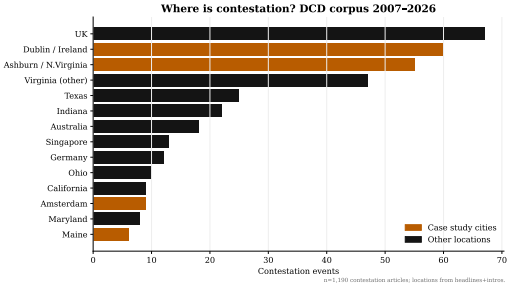
Cluster type	Example terms
Energy / grid	MW, capacity, power, grid
Investment / capex	$B, funding, lease, REIT
Planning / planning	council, rezone, permit
Contestation	moratorium, oppose, sue
Cooling / efficiency	PUE, liquid, immersion
Geography	Virginia, Ireland, Singapore

Results running — will appear in updated slides.

`data/processed/wp4_dcd/bertopic_topics.csv`

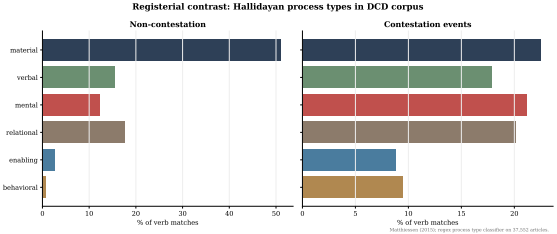
The Arena in Data





37,552 articles | 1,073 contestation events | peak 2025. **Headlines + intro text only** — full articles via screenshot/OCR pipeline. More coming.

The Register Shift



Matthiessen (2015) registerial cartography — verb process types

Type	Contest	Non
Behavioral	23.1%	1.6%
Enabling	16.5%	3.1%
Material	33.7%	66.8%

Doing → governing + contesting

The governance gap visible in the verb register.

Headlines + intros only.

What the Cartography Enables

Three research agendas the Cartalog makes tractable:

Agenda	Central question
Regional innovation systems	How do hyperscale clusters reshape local labour, energy, and planning systems?
Sectoral transformation	How does hyperscaler vertical integration close sectoral entry points for new firms?
Policy sovereignty & energy	What governance mix sustains compute capacity without fossil lock-in or geopolitical capture?

Each operationalized via CITC (Berente et al. 2019): sampling → synchronic → diachronic → lexical framing.

Full agenda + cases: appendix.

Three Contributions

Cartalog of Actors

Who is in the datacentering arena?

40,565 DCD articles → NER → actor typology: hyperscalers, planners, communities, courts, utilities

Geocoded, dated, open (CC BY 4.0)

Cartalog of Events

What happens in the arena?

1,036 contestation events → event type, location, date, actors involved

Register shift: *doing* → *enabling + contesting*

Peak: 2025 (214 events)

Methodological Toolkit

How to study the arena?

Arenas of Transformation (after Jørgensen 2012)

+ Situational maps (Clarke)
+ CITC (Berente et al. 2019)
+ Registerial cartography (Matthiessen 2015)

Replicable. Open pipelines.

Building the Cartography Together

Why open is theoretical necessity

CAS-like phenomena are too distributed, too opaque for any single team.

What we share

- Facility index (CC BY 4.0)
- NLP / CV / network pipelines (MIT)
- DCD contestation corpus
- Register + NER analysis scripts
- `github.com/babajideowoyele/datacentering-cartography`
- **Cartalog UI** — live actor network, topic map, spatial map

Concrete asks

- Local planning records + disputes
- Non-English corpus (NL, DE, FR, IE)
- Company registry / REIT data access
- Replication in other jurisdictions
- Methodological critique

*"What would innovation studies look like if we treated data infrastructure as **foundational** rather than peripheral?"*

Key References i

Theory & framework

Jørgensen (2012) — Arenas of development. *Research Policy* 41(6), 996–1010

Clarke (2005) — *Situational Analysis*. SAGE

Marres (2015) — Why map issues? *ST&HV* 40(5), 655–686

Berente et al. (2019) — CITC. *ISR* 30(1), 50–64

Miranda et al. (2016) — Sensemaking. *MISQ* 40(2), 413–444

Matthiessen (2015) — Registerial cartography. *Funct. Ling.* 2(9)

Critical DC studies

Edwards, Cooper & Hogan (2025) — Making of critical DC studies. *Convergence* 31(2)

Monstadt & Saltzman (2025) — Governing DC footprint. *IJURR* 49, 757–778

Young (2026) — DCs and local governance. *Oxford Open Energy*

Energy & infrastructure

Masanet et al. (2020) — Recalibrating DC energy use. *Science* 367(6481)

IEA (2024) — Electricity 2024. Paris: IEA

IEA (2025) — Energy and AI. Paris: IEA

Data & corpus

DCD corpus — *DatacenterDynamics* crawl
40,824 articles, 2006–2026
Headlines + intro text

Geels (2002) — Multi-level perspective. *Res. Policy* 31(8–9)

Empirical sources

PJM (Dec 2025) — Capacity auction results

CreditSights (Feb 2026) — Hyperscaler capex

Moorhouse (2026) — DC vs. megaprojects capex

`github.com/babajideowoyele/datacentering-carto`

Three Research Agendas — Full

Agenda	Central question	Cases
Regional Innovation Systems	How do hyperscale clusters reshape local productivity, labour, energy, property?	Loudoun (US), Luleå (SE), Dublin (IE), Hyderabad (IN)
Sectoral Transformation	How does hyperscaler vertical integration reshape sectoral systems and startup access?	AWS, Google, Meta, Microsoft platform dependency
Policy Sovereignty & Energy	What policy mix sustains sovereign compute without fossil lock-in or geopolitical dependency?	UK AI Growth Zones, EU AI Factories, France, Saudi HUMAIN, UAE Stargate

Each agenda operationalized via the CITC cycle (Berente et al. 2019): sampling → synchronic → diachronic → lexical framing.

Q: Isn’t this STS dressed up as innovation studies?

Live: “Fair point — I adapt Jørgensen’s *arenas of development* (RP 2012) into *arenas of transformation* because the question isn’t how DCs get built but what they do to the systems around them: energy grids, planning regimes, labour markets. The *transformation* framing is also consistent with transitions vocabulary your audience knows.”

The methodological spine — CITC (Berente, Seidel & Safadi 2019; Miranda et al. 2022) — comes from *MIS Quarterly* and *ISR*, not from STS. The agenda is testable, not hermeneutic.

Q: Where's the empirical contribution?

Live: "The ECR paper is the agenda. The corpus analysis is the first empirical instantiation."

From 37,552 DCD articles:

Analysis	Status	Finding
EDA geography	Complete	US/Europe split; peak 2025
Register analysis	Complete	Behavioral +21.5%, enabling +13.4%
BERTopic topics	Running	Full corpus topic clusters
city2graph	Seed run	3 cities; scales via GLiNER NER

Analysis on headlines + intros. Full article text: screenshot/OCR pipeline coming.

Q: Why Loudoun, Luleå, Dublin, Hyderabad?

Variation on three dimensions:

Case	Regulatory	Energy	Trajectory
Loudoun	Common law / NIMBY	Gas + solar	Mature US cluster
Luleå	Swedish planning	98% hydro	Renewable-led
Dublin	Irish planning	Mixed	FDI platform
Hyderabad	Indian regulatory	Expanding	Emerging hub

Q: What are the register verb families?

Following Matthiessen (2015) Hallidayan process types — applied to DCD headlines + intros:

Process type	Field of activity	Top verbs in corpus
Material	Doing / building	secures, builds, launches, withdraws, rejects
Verbal	Reporting	says, announces, claims, warns, demands
Mental	Exploring / sharing	opposes, concerns, challenges, supports
Relational	Expounding	is, becomes, remains, includes
Enabling	Enabling	bans, permits, rezones, mandates, regulates
Behavioral	Contesting	protests, campaigns, sues, appeals, petitions

Contestation = shift from material/relational to behavioral/enabling.

Datacentering × Transitions Research Agenda i

Tensions and paradoxes surfaced by the *arena of transformation* framing, mapped onto the nine themes of the updated transitions research agenda (Geels et al. 2024, *Environ. Innov. Soc. Transitions*).

Datacentering × Transitions Research Agenda ii

Theme	Tension / Paradox in Datacentering
i. Understanding transitions Geels, Fünfschilling	DCs simultaneously *enable* digital transitions and *block* energy transitions. Same infrastructure; opposite systemic effects. Standard transition frameworks assume one trajectory — datacentering has at least two.
ii. Power & politics Avelino, Kern	Hyperscalers negotiate bilaterally with grid operators at national scale yet face municipal veto at planning level. Power is scale-incommensurable — no single arena contains it.
iii. Governing transitions Nykvist, Kivimaa	AI infrastructure strategy formed nationally (Stargate, EU AI Act); planning permission granted locally. Governance most fragmented precisely where infrastructure is most cross-scalar.
iv. Civil society & movements Hess, Rohracher	"Clean cloud" is marketed globally; noise, heat-island, and water stress are locally borne. Spatial mismatch of benefit and burden is structural, not incidental.
v. Organizations & industries Markard, Wells	Three hyperscalers control >50% of global capacity while claiming to enable a distributed, competitive digital economy. Sector consolidation and democratisation discourse co-evolve.
vi. Practice & everyday life McMeekin, Hyysalo et al.	"Dematerialisation" (cloud = less hardware) coexists with the fastest physical infrastructure build on record. Jevons paradox at the cultural level: invisibility amplifies consumption.
vii. Geography of transi-	Global cloud routes through three postcodes — Ashburn, AMS-IX, Slough. Geographic concentration *increases* as scale

Q: Why CITC and not CSS more broadly?

CITC's lexical-framing requirement is what makes the output theoretical, not just descriptive.

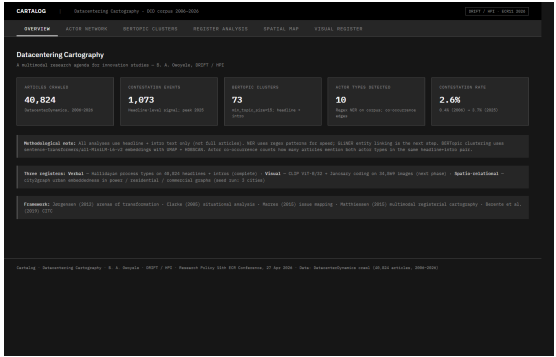
That's the discipline that protects against the most common reviewer objection:

> *"This is interesting, but what's the theoretical contribution?"*

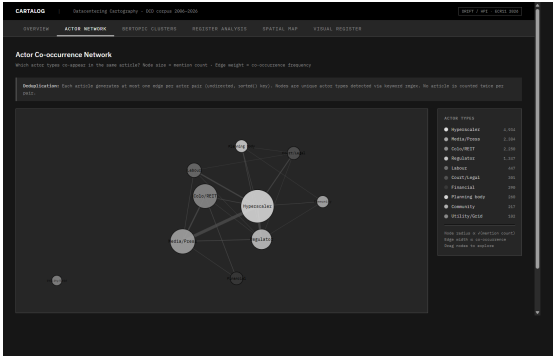
CITC explicitly distinguishes patterns from contributions. The test for RP.

Cartalog — Live Dashboard

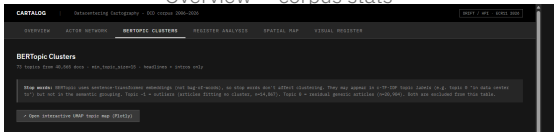
babajideowoyele.github.io/datacentering-cartography/docs/cartalog/



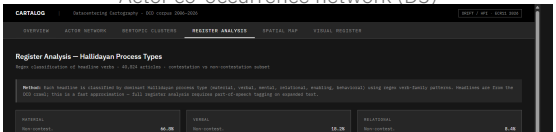
Overview — corpus stats



Actor co-occurrence network (D3)



Overview — corpus stats



Actor co-occurrence network (D3)

Visual Register Analysis — Coming Next i

34,869 image URLs in the DCD corpus. Visual register = what imagery *says* about the arena — who is shown, how, in what settings.

Pipeline (next phase):

1. Download images from DCD CDN
2. CLIP ViT-B/32 embeddings → cluster
3. Gemma 4 (Ollama, 24GB GPU) → caption
4. Code against Jancsary et al. scheme
5. Visual Register Index (VRI) per year / actor

Research question:

Does visual register shift as contestation rises? Do hyperscaler images show *nature / community* as contestation increases, and if so, when does that shift begin?

Jancsary et al. — Ideational metafunction

Aspect	Relevance
Content: Participants, attributes, processes, settings	Category manifestations; central truth claims
Conjunction: Narrative / conceptual relations	Internal institutional "architecture"

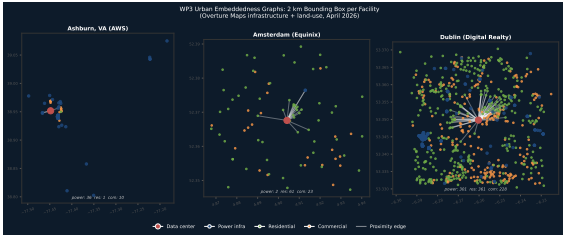
Interpersonal metafunction

Aspect	Relevance
Contact, social distance, vertical/horizontal angle	Power relations; actorhood; institutional gaze
Coding orientations: naturalistic, sensory, abstract	Aesthetic / emotional legitimation

Visual Register Analysis — Coming Next ii

Benchmark content coding (Jancsary et al.): Male adults 35% · Female adults 14% · Non-descript settings 47% · Production facilities 12% · Natural spaces 12% · Office settings 8% (n=1,023 reference images). DCD corpus will be coded against this scheme to produce a Visual Register Index (VRI) tracking legitimation strategy over time.

City2Graph: Urban Embeddedness



From corpus cities → urban graphs

GLiNER NER extracts city names
→ city2graph pulls Overture Maps
→ graph: DC node ↔ power, residential, commercial

	Res. zones	Power
Ashburn	1	36
Amsterdam	61	2
Dublin	361	6

Spatial embeddedness determines governance counterparty availability.

Seed run: 3 cities. Scales via GLiNER NER on full 40,824-article corpus.